HDOC Day-43 - C Programming

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Problem Statement

Write a menu-driven C program to perform the following operations using separate functions:

1. Find the longest word in a sentence.
2. Check if one string is a rotation of another.
3. Exit.

Header Files Used: stdio.h, string.h

C Program

```
#include <stdio.h>
#include <string.h>

void findLongestWord(char sentence[])
{
    char temp[200];
    char longest[100] = "";
    char *word;

    strcpy(temp, sentence);
    word = strtok(temp, " ");

    while (word != NULL)
    {
        if (strlen(word) > strlen(longest))
            strcpy(longest, word);

        word = strtok(NULL, " ");
    }

    printf("Longest word: %s\n", longest);
}

int checkRotation(char str1[], char str2[])
{
    char temp[200];

    if (strlen(str1) != strlen(str2))
        return 0;

    strcpy(temp, str1);
    strcat(temp, str1);

    if (strstr(temp, str2) != NULL)
        return 1;
    else
        return 0;
}

int main()
{
    int choice;
    char sentence[200];
    char str1[100], str2[100];

    do
    {
        printf("\n----- MENU -----\n");
        printf("1. Find longest word in a sentence\n");
        printf("2. Check string rotation\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
```

```
        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                printf("Enter a sentence: ");
                fgets(sentence, sizeof(sentence), stdin);
                sentence[strcspn(sentence, "\n")] = '\0';

                findLongestWord(sentence);
                break;

            case 2:
                printf("Enter first string: ");
                fgets(str1, sizeof(str1), stdin);
                str1[strcspn(str1, "\n")] = '\0';

                printf("Enter second string: ");
                fgets(str2, sizeof(str2), stdin);
                str2[strcspn(str2, "\n")] = '\0';

                if (checkRotation(str1, str2))
                    printf("The strings are rotations of each other.\n");
                else
                    printf("The strings are not rotations of each other.\n");
                break;

            case 3:
                printf("Program exited successfully.\n");
                break;

            default:
                printf("Invalid choice.\n");
        }

    } while (choice != 3);

    return 0;
}
```

Program Output

